AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

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Presentation Outline

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Problem Statement

Dead spacecrafts accumulating as space debris creating a high-risk environment in orbit.

Problem Understanding

- **Who experiences this problem?:** Government Space agencies and Defence Organisations(ISRO, NASA, DRDO. . .); Commercial satellites and satellite constellation operators Reliance JIO, TATA Sky, Starlink...)
- **What causes the problem?:** Spacecraft/Probes that died in orbit left unchecked causing collisions with other debris i.e., Kessler Syndrome
- **How does it affect users or stakeholders?:** Active satellites and spacecrafts are in under threat of aggressive contact with debris particles which will hinder or completely disrupt it's functions. Everyone relying on that craft for any service such as GPS, telecommunication and The Internet will be completely unable to access those services henceforth. Only way to fix the service would be to abandon the original craft and send another craft to take over. In the end it just creates more dibri and the cycle goes on.

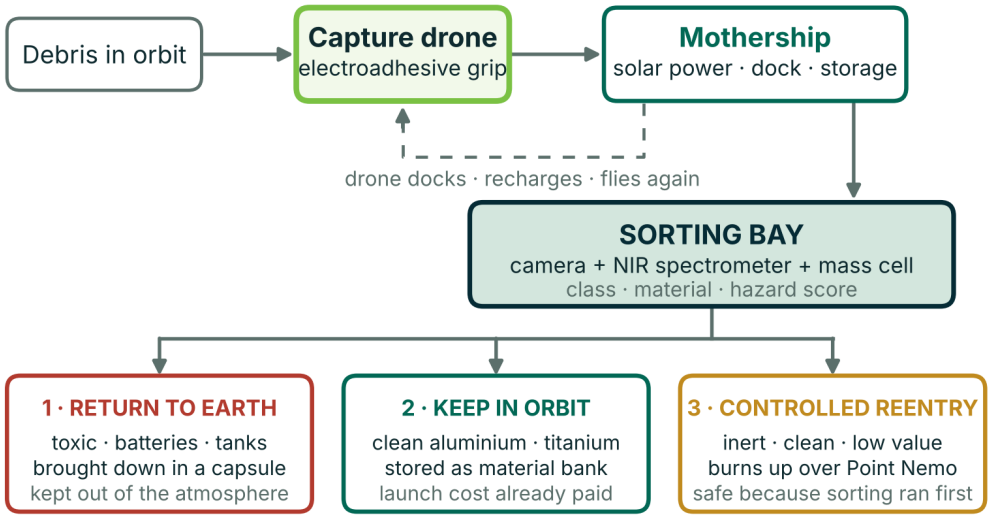
Existing Gaps & Challenges

- **Existing gap or limitation 1 :**In ADRAS-J (Japan) - Debris was identified as a target and only the chosen target was studied. In second iteration to be launched in 2027 the ADRAS-J2 will take action but only on the target studied by ADARS-J.
- **Existing gap or limitation 2 :**DFSM (Debris Free Space Mission) - Announced by ISRO in the press conference of April 16 ,2024 by S.Somanath (ISRO Chairman) on the occasion of the inaugural opening plenary of the 42nd Annual Meeting of the Inter-Agency Space Debris Co-ordination Committee (IADC) in Bengaluru, India. The core idea is to have debris free Indian space actors by 2030. Post the conference ISRO as yet to reveal specific steps as taken to achieve the goal.
- **Main challenge that remains unsolved:**All the solutions so far have been targeting single pieces of debris and has been almost completely neglecting the smaller pieces of debris that are arguably more dangerous.

Our Proposed Solution

- **Core idea or product:** PROJECT T.A.N.D.A.V. (Technology for Assimilated, Neglected Debris Accumulation Vehicle). A mothership with multiple deployable drones with 2 major functions. Collect debris from orbit to depositing it in the mothership storage and collect from mothership storage to drop above the collection point.
- **How the solution works at a high level :**Project T.A.N.D.A.V. targets all forms of unclaimed space debris with agreement from the nation that it is a propert of leaglly.
- **Main value delivered to users**Reduced risk of damage to service providing crafts.

Technical Approach



Every mission flying today has only exit 3. The sorting bay is what creates exits 1 and 2.

Reference: https://www.anthropic.com

Tech Stack

Frontend: ReactNative(for Public)

Backend: C,C+

Database: Space Track

AI/ML, APIs, or Hardware: Convolution Neural Network,ESP-32

Innovation & Uniqueness

- **What is new or different?** Unlike existing and announced solutions, Project T.A.N.D.A.V. doesn't just target one piece of large debris but multiple pieces of different-sized sets.

- **Why is this approach suited to the problem?** Space debris is in quantities of millions in orbit at this very moment, and with current solutions of only tackling just a fraction of the debris that is mostly larger in size, the smaller particles are usually left out, increasing the danger of these "space bullets" that have high velocities, and hence if they collide with any other object, they can have a huge impact.

- **What is the key differentiator?** The key differentiators between Project T.A.N.D.A.V. and equivalent projects are multi-target automation and AI tracking software with public participation.
  - Multi-target Automation
  - AI Tracking with Public Participation

Feasibility & Scalability

- **Feasibility:** Nothing here depends on inventing new technology. Capture, docking, and propulsion are things spacecraft already do today, we're arranging them differently, not reinventing them. The only genuinely new step is sorting debris before disposal, and that's the part we've already built and run on a table-top rig. Everything else is engineering work we know how to do.

- **Scalability:** We're starting small, just Indian-owned debris, since no other country's permission is needed for that. Once it works, the same system can handle more objects at once and expand to busier orbit zones, lining up with ISRO's existing debris-free mission plan.

- **Risks:** Debris that's been in space a long time might confuse our sensors since its surface wears out over the years , we'd recheck it against records if that happens. Also, we can only bring back a limited weight per trip, so only the genuinely hazardous stuff comes back to Earth.

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: ISRO—Recovers PSLV upper stages to reuse materials for upcoming projects
- Use case 2: NASA—Recovers Vanguard 1 to learn from its past and find a different generation of science used.
- Use case 3: ISS—Cleaning up the KOSMOS 1408 fragments so that future crew doesn't have to be forced into shelter again.

Impacts and Benefits

- **Benefit to the target users:** Satellite operators get a clearer orbit to work in — fewer dodging manoeuvres, less fuel wasted avoiding junk. Space agencies get a way to clean up their own old objects without needing anyone else's permission.

- **Social, economic, or environmental impact:** The toxic parts come home instead of being burned into the upper atmosphere, so we stop trading an orbit problem for an air-quality one. Clean metal stays up there as usable material instead of being destroyed after we already paid to launch it.

- **Expected measurable outcome**More objects removed per year than today's one-per-mission missions manage, hazardous material brought back rather than burned, metal banked in orbit, and zero captures that create new debris — plus the first real data on what space junk is actually made of.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 12	Target 1	Responsible consumption and production	Promoting responsible management of space waste and exploring potential material recovery
SDG 14	Target 2	Life below water	Assessing environmental consequences of debris disposal and preventing marine pollution

Individual Member Contribution

Member	Contribution or responsibility
Aadrika Bhardwaj	Team Leader
R Samyukta	Sustainable development goals
Parthiv PC	Existing Gaps and Challenges
Abinash S	Feasibility and Scalability

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